Politecnico di Milano

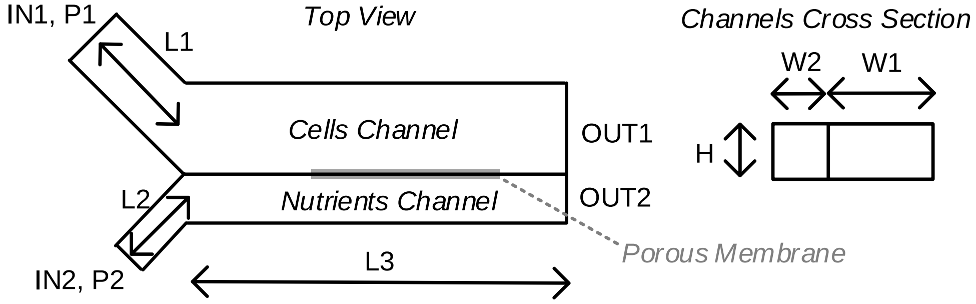
Biochip A.Y. 2019/2020 - M. Carminati

On-Line Exam of 3.07.2020

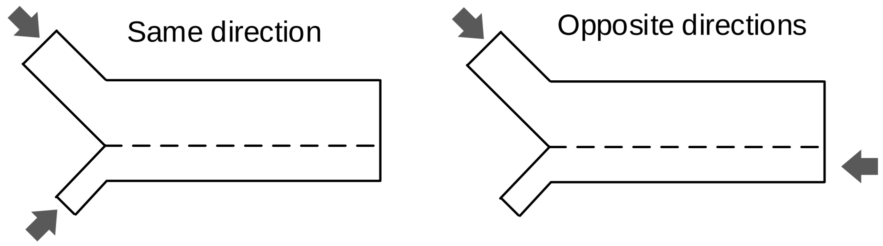
Test duration: 2 hours – No support material admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

As usual the exam is composed of two parts: question 1-7 focus on the first system, while questions 8-12 on the second one.

1. Consider the following microfluidic system composed of two channels (one for cells, and one for nutrients) separated by a porous membrane that allows the passage of chemical species: H = W2 = 50μm, W1 = 200μm, L1 = 3mm, L2 = 2mm, L3 = 10mm. Find the electrical equivalent of the fluidic circuit.

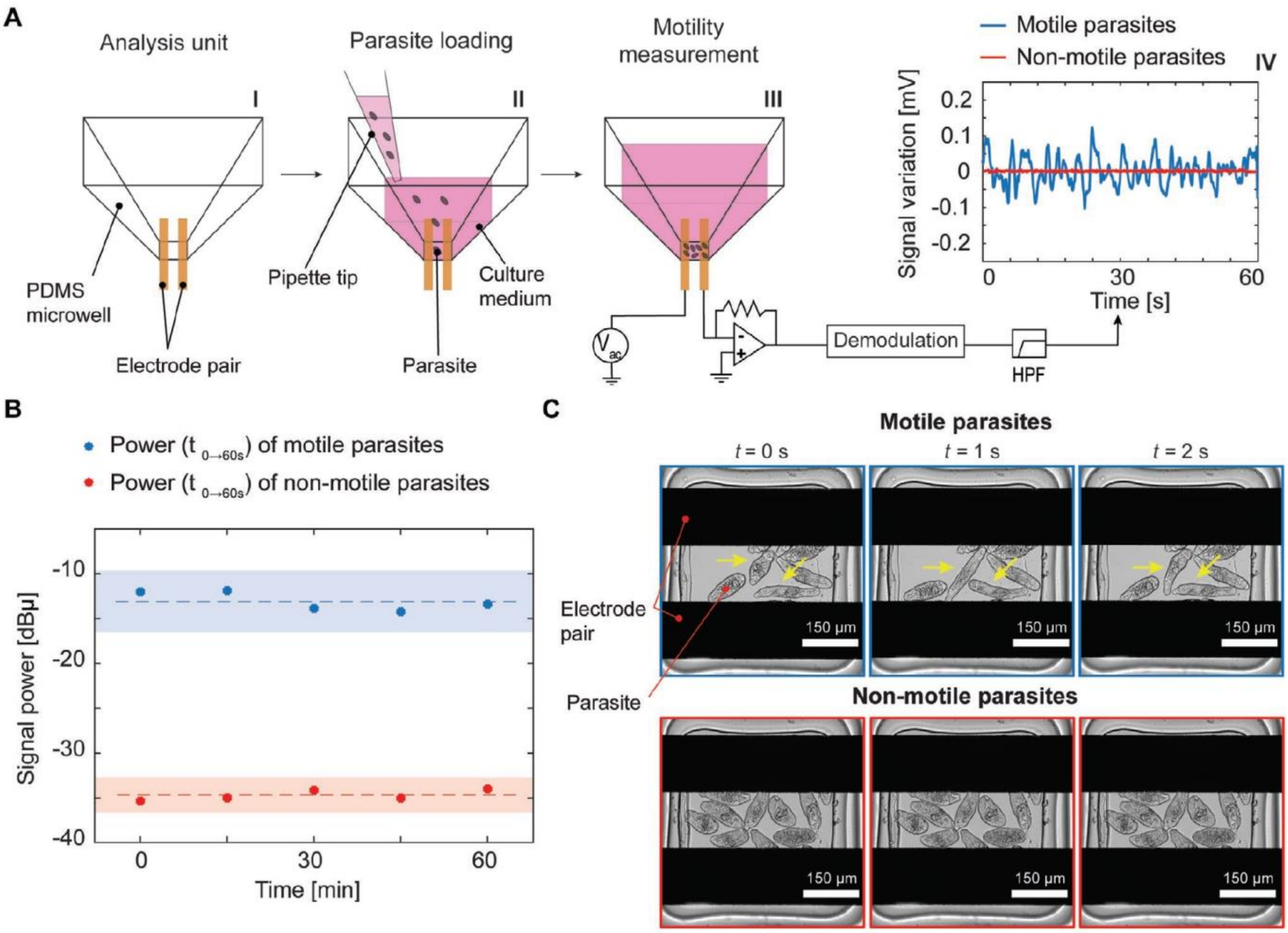


2. If P1 = 1.5 atm and P2 = 2 atm, $P_{OUT1} = P_{OUT2}$ = Pair = 1 atm = 101kPa, find the velocity of the fluid in each channel.
3. Considering DNA fragments of 1μm gyration radius travelling in L3 region at a velocity of 1mm/s, what is the drag force experienced by this particles?
4. Assuming that that DNA fragment (coiled in a globular shape) has a length of 10kpb and it is driven by electrophoresis with a couple electrodes separated by L3, what would be the voltage difference required to achieve a velocity of 1mm/s? Draw the polarity of the voltage drop.
5. If the membrane separation allows the passage of molecules with a diffusion coefficient larger than $4 \cdot 10^{-11} m^2/s$, what is the maximum diameter of the molecule that can pass between the channels?
6. Now assuming that the membrane between the two channels is removed, compares the two strategies shown in the picture from the point of view of enhancement of mixing between fluids.



7. Describe the fabrication of this device in two cases: (i) made of PDMS and (ii) made of PMMA (rigid plastic).
8. Design a <u>digital microfluidic system</u> performing the same operation of this device.

9. Now consider the system shown in the picture targeting the measurement of the motility of parasites by means of impedance. Draw the scheme of the "Demodulation" block and choose the bandwidth of this bock in order to track the motility with a time resolution of 1 second.



10. If parasites are in PBS solution, the area of gold electrodes is 2000µm^2, the solution resistance equal to 10kΩ and the parasitic capacitance due to the connections is 1pF, find the impedance equivalent model and plot the Bode diagrams (magnitude and phase).
11. Given the impedance spectrum of the previous point and a feedback resistor of the TIA of 100kΩ, choose the best sensing frequency to measure motility with impedance tracking.
12. Find the maximum voltage noise (spectral density) of the signal generator V_{AC} that produces the same input-referred current noise of the TIA feedback resistor in the white region.